

# R TUTORIAL - Signal Processing Journal Club

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# Outline

- 1 What is R? Why/Why not R? Practical Issues.
  - What exactly is this R thing?
  - The Good, The Bad & the Ugly (Sides of R)
  - Practical Issues
- 2 Introductory Examples & the R Paradigm
- 3 Markdown
  - What is a Markdown file?
  - Markdown Example
- 4 Plotting and Graphics with R
  - Base Plotting System
  - Lattice Plotting System
  - Grammar of Graphics (ggplot2) Plotting System
  - External Packages
- 5 Machine Learning & Statistical Learning with R
  - Useful Packages and Examples
- 6 Miscellaneous Stuff
  - Interactive Applications
  - API Interaction
  - And more...

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# What is R? Why/Why not R? Practical Issues.

## What exactly is this R thing?

R is a language and environment for statistical computing and graphics. It provides a wide variety of statistical (linear and nonlinear modeling, classical statistical tests, time-series analysis, classification, clustering, ...) and graphical techniques, and is highly extensible.



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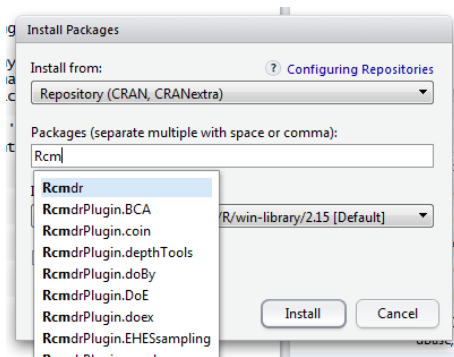
## 6 Miscellaneous Stuff

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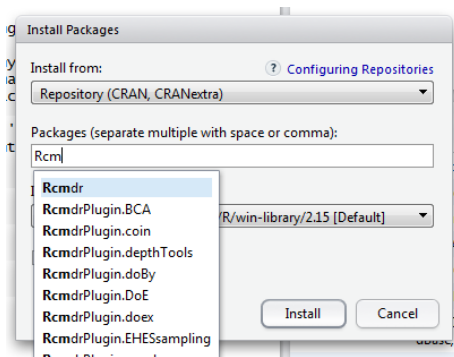
### ■ Free (Speech & Sandwich)



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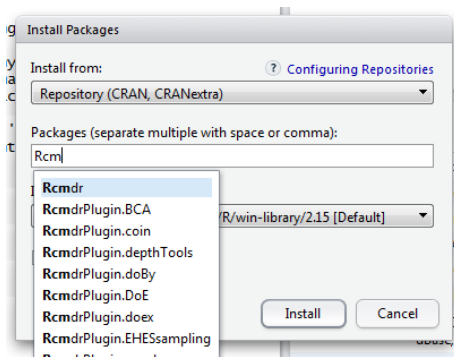
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## The Good, The Bad & the Ugly (Sides of R)

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- Most comprehensive statistical analysis environment
- Maintained by many statisticians, scientists

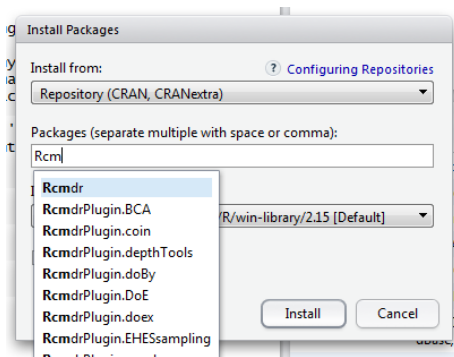




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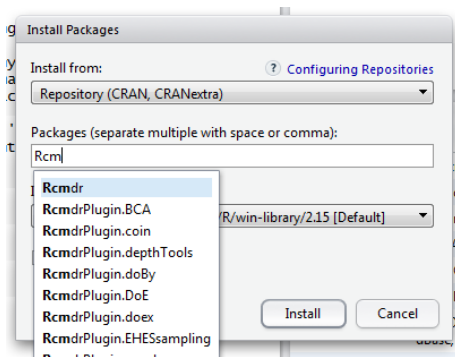
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- Outstanding graphical capabilities



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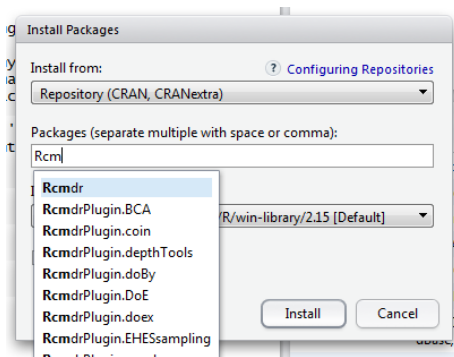
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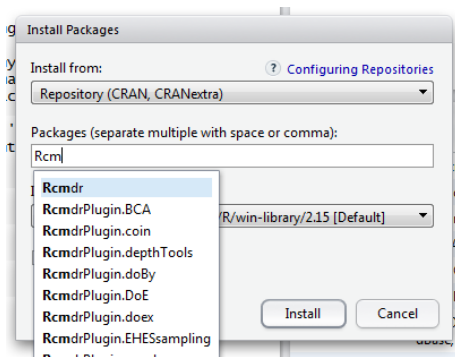
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- Free (Speech & Sandwich)
- Most comprehensive statistical analysis environment
- Maintained by many statisticians, scientists
- Outstanding graphical capabilities
- Almost 5000 packages
- Cross-platform
- Active user group



# What is R? Why/Why not R? Practical Issues.

## The Good, The Bad & the Ugly (Sides of R)

### ■ ☹ Learning Curve

```
<?xml version="1.0" encoding="UTF-8"?>
```

although it does not require it.

The [XML](#) package provides general facilities for reading and writing XML documents within R. A description of the facilities & details and examples. Package [StatDataML](#) on CRAN is one example building on [XML](#).

NB: [XML](#) is available as a binary package for Windows, normally from the 'CRAN extras' repository (which is selected by default).

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2 Spreadsheets

- [Variations on read.table](#)
- [Fixed-width-format files](#)
- [Data Interchange Format \(DIF\)](#)
- [Using scan directly](#)
- [Re-shaping data](#)
- [Flat contingency tables](#)

In [Export to text files](#) we saw a number of variations on the format of a spreadsheet-like text file, in which the data are presented in a rectangular grid.

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#### 2.1 Variations on read.table

The function `read.table` is the most convenient way to read in a rectangular grid of data. Because of the many possibilities, then, because that `read.table` is an inefficient way to read in very large numerical matrices: see [scan](#) below.

Some of the issues to consider are:

##### 1. Encoding

If the file contains non-ASCII character fields, ensure that it is read in the correct encoding. This is mainly an issue for `read.table("file.dat", fileEncoding="latin1")`.

Note that this will work in any locale which can represent Latin-1 strings, but not many Greek/Russian/Chinese/Japanese.

##### 2. Header line

We recommend that you specify the `header` argument explicitly. Conventionally the header line has entries only for the columns presented with a file that has a (possibly empty) header field for the row labels, read it in by something like

```
read.table("file.dat", header = TRUE, row.names = 1)
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Column names can be given explicitly via the `col.names`: explicit names override the header line (if present).

##### 3. Separator

Normally looking at the file will determine the field separator to be used, but with white-space separated files there may be "v". Note that the choice of separator affects the input of quoted strings.

If you have a tab-delimited file containing empty fields be sure to use `sep = "\t"`.

##### 4. Quoting

By default character strings can be quoted by either `"` or `'`, and in each case all the characters up to a matching quote as argument. For `sep = "\t"` the default is changed to quote = "".

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- ☹ Documentation style

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- NaN (not a number), NA (not available)

# Assignment, Lists, Arrays etc.

---

## introduction.R, control\_structures.R

---

- Assignment

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---

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---

- Assignment
- Indexing

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- Assignment
- Indexing
- Arrays
- List
- Control structures

# Data Frames

---

## data\_frames.R

---

- Creating DFs

# Data Frames

---

## `data_frames.R`

---

- Creating DFs
- Using DFs



# Data Frames

---

## `data_frames.R`

---

- Creating DFs
- Using DFs
- Aggregation

# Data Frames

---

## `data_frames.R`

---

- Creating DFs
- Using DFs
- Aggregation
- Reshaping

# 'apply' functions

---

## apply\_functions.R

---

- lapply

# ‘apply’ functions

---

## apply\_functions.R

---

- lapply
- sapply

# ‘apply’ functions

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## apply\_functions.R

---

- lapply
- sapply
- tapply

# Statistical Tools

- Prefix d for density value (pdf)

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- Output of a standalone HTML file (with images embedded directly in the file) that is easy to share using email, Dropbox, or by deploying to a web server.

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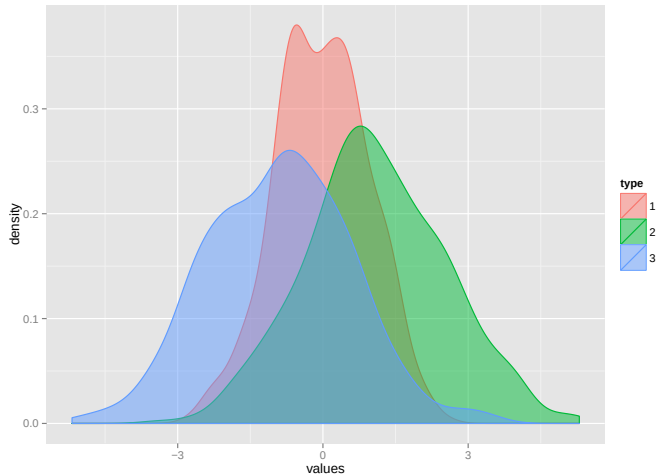
- Easy creation of web reports from R that can be automatically regenerated whenever underlying code or data changes.
- A highly accessible syntax (markdown) which lower the barriers to entry for reproducible research.
- Output of a standalone HTML file (with images embedded directly in the file) that is easy to share using email, Dropbox, or by deploying to a web server.
- Support for publishing dynamic and interactive web content.

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# Markdown Example



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# Base Plotting System

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**base\_introduction.R, base\_exploratory.R, maps\_example.R**

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Idea: High-level graphing function followed by low-level modifying functions + default parameters

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Most common high-level ones: *plot*, *hist*, *barplot*, *boxplot*, *curve*

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Most common low-level ones: *lines*, *text*, *axis*, *points*, *title*, *mtext*

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Most common low-level ones: *lines*, *text*, *axis*, *points*, *title*, *mtext*

Global graphics parameters: *par()*

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# Lattice

Useful for *trellis graphs*, i.e., graphs that display a variable or the relationship between variables, conditioned on one or more other variables.

*formula* argument:  $y \sim x \mid p * r$

where

y and x are primary variables

p and r are conditioning variables



# Lattice

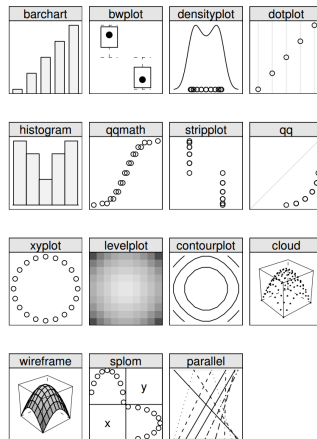
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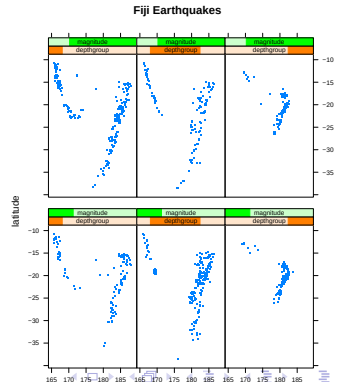
p and r are conditioning variables



# Lattice Introductory Example

## `lattice_introduction.R`

Plot Fiji earthquakes, i.e., plot latitude vs. longitude conditioned on depth and magnitude.



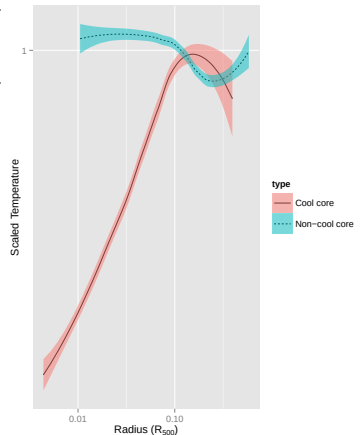
# Outline

- 1 What is R? Why/Why not R? Practical Issues.
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# Grammar of Graphics (ggplot2)

## ggplot2\_introduction.R

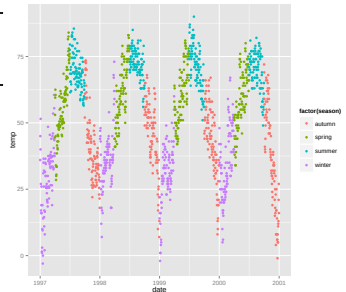
Plot galaxy clusters info with error intervals.



# Grammar of Graphics (ggplot2)

## ggplot2\_example2.R

Several simple plot examples on the same dataset.

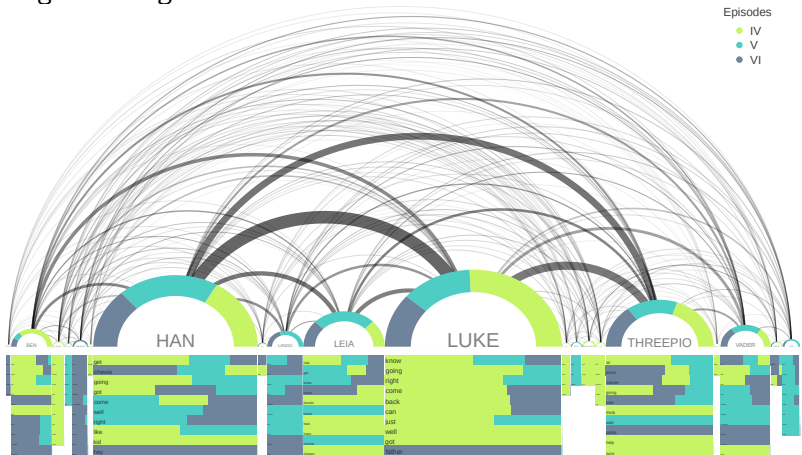


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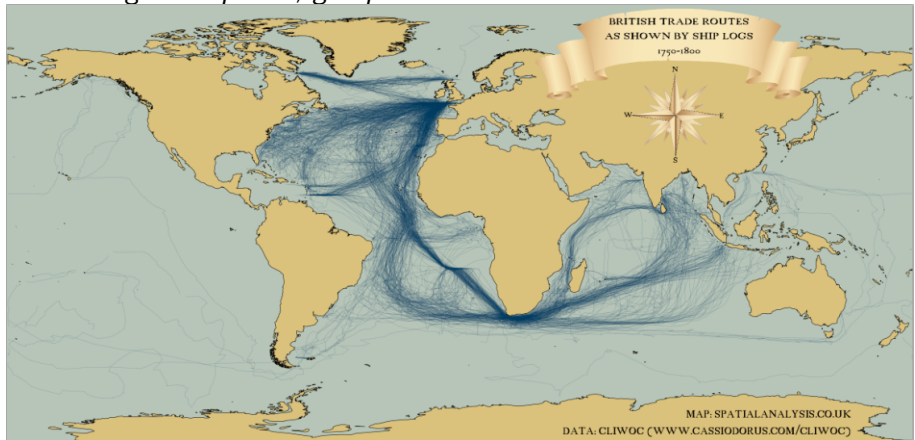
# External Packages

Package: *arcdiagram*



# External Packages

Packages: *maptools*, *geosphere*





# Graphing Roadmap

| Task                 | Graphing Tool     |
|----------------------|-------------------|
| Exploratory Analysis | base              |
| Simple 2D Plots      | base              |
| Quick Plotting       | base              |
| Trellis Graphs       | lattice           |
| Formal Plotting      | ggplot2           |
| Beautiful (?) Plots  | ggplot2           |
| Unconventional Plots | external packages |

Graphing Roadmap

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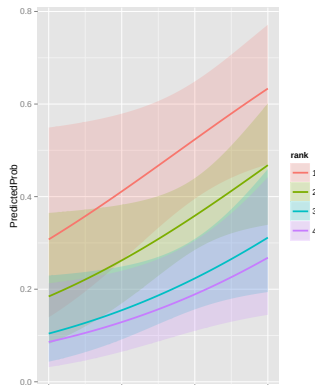
Some useful packages:

- **caret** : General machine learning
- **nnet** : Neural networks
- **randomForest** : Decision trees/random forests
- **bigrf** : Random forests for large datasets
- **gbm** : Gradient boosting
- **lars**, **lasso2**, **glmnet** : Regularized methods
- **e1071**, **kernlab** : SVMs and kernel methods
- **rgp**, **rgenoud** : Genetic algorithms

# Logistic Regression Example

## `logistic_regression.R`

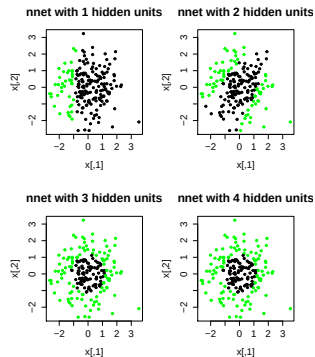
Predict graduate school admission probabilities using logistic regression.



# Neural Network Example

## neural\_net.R

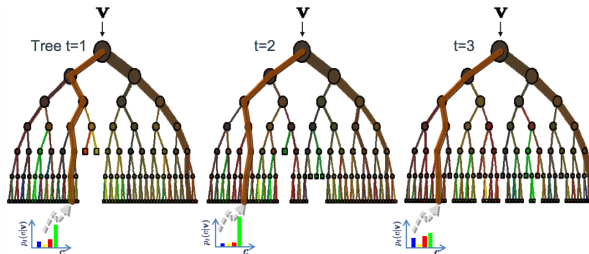
Binary classification using a very simple neural network classifier.



# Random Forest Example

## `random_forest.R`

Binary classification using a very simple random forest classifier.



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## Interactive Plotting and Documentation

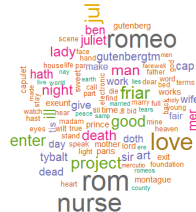
Conduct document analysis on Shakespeare's plays in an interactive way using the **shiny** package.

Choose a book:  
Romeo and Juliet

Change

Minimum Frequency:  
1 15 50

Maximum Number of Words:  
1 100 300





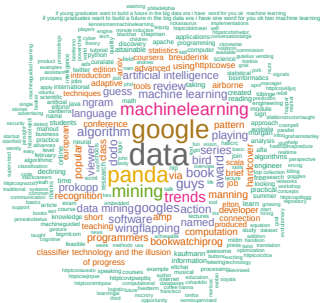
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Interactive Applications  
API Interaction  
And more...

## API Interaction

Data mining on Twitter using twitteR package.



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# Miscellaneous Stuff

And more...

One can

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